

Coal Mine Compliance Monitor

Who uses this system, and how it flows — a plain-English guide (SIH26024)

The system fuses three kinds of evidence — statutory documents, field inspections, and satellite monitoring — into one continuously-updated compliance score per mine. When a mine's score drops too low, the system opens a case, and four kinds of people move it toward resolution.

The Four Roles

Ministry Admin

The top-level view — sees the whole country.

- Sees every mine, in every state, all at once.
- Views ministry-wide reports and statistics.
- The audience for the overall governance picture.

DGMS Officer

The regional regulator — the one who actually acts on cases.

- Only sees mines in their own state/jurisdiction.
- Reviews findings on a mine's compliance score.
- Assigns a case to a Field Inspector to go check it in person.
- Verifies the inspector's evidence and formally closes the case.

Mine Manager

Represents one specific mine — the operator's side.

- Sees only their own mine's score, documents, and cases.
- Uploads compliance documents (clearances, permits, certificates).
- Submits remediation evidence once a problem has been fixed.

Field Inspector

Boots on the ground — the one who checks reality.

- Sees only the cases assigned to them.
- Fills out a pass/fail safety checklist on site.
- Captures GPS location and a photo as evidence.
- Can save an inspection offline and sync it later once back online.

One rule ties all four together: every action is checked on the server by role, not just hidden in the screen — an inspector can never open a case that isn't theirs, and a manager can never see another mine's data.

The Compliance Score

Every mine gets one score out of 100, built from four weighted parts. A low score in any one part pulls the overall score down — nothing is hidden inside an opaque “AI decided” number.

Statutory Documents	30%	Are clearances, permits and certificates valid and not expired?
Safety / Field Inspection	35%	Do on-site checklist results show real safety compliance?
Environmental / Satellite	25%	Any unexplained vegetation loss or boundary encroachment?
Operational / Other	10%	Other configured indicators (e.g. statutory returns filed).

GREEN	80 – 100 — Compliant — no action needed.
YELLOW	60 – 79 — Some issues — worth watching.
RED	below 60 — A case is opened automatically, ready for a DGMS officer to assign.

How a Case Moves, Start to Finish

DETECTED	TRIAGED	ASSIGNED	REMIATI ON	EVIDENCE SUBMITTED	VERIFIED	CLOSED
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- 1. Detected → Triaged** — the score engine opens a case by itself the moment a mine turns RED, listing exactly which findings caused it.
- 2. Assigned** — a DGMS Officer reviews the case and assigns it to a Field Inspector.
- 3. In Remediation** — the inspector visits the mine; the mine manager can also submit fixes on their end.
- 4. Evidence Submitted** — the inspector's checklist, GPS location and photo are attached to the case automatically.
- 5. Verified → Closed** — the DGMS Officer checks the evidence and formally closes the case with a reason. The mine's score recalculates immediately.

In one sentence: documents, inspections and satellite signals all feed one honest score per mine, and every problem is tracked from detection to a verified, closed resolution — not just a map, a chatbot answer, or a report that nobody has to act on.